simplificationoutput

April 23, 2026

[52]:
```
import warnings
warnings.filterwarnings("ignore")

from pathlib import Path
import subprocess
import sys
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.linear_model import LinearRegression, LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import (
    roc_auc_score,
    average_precision_score,
    brier_score_loss,
)
from sklearn.calibration import CalibratedClassifierCV, calibration_curve

try:
    from lifelines import KaplanMeierFitter, CoxPHFitter
    from lifelines.utils import concordance_index
except ModuleNotFoundError:
    print("Installing lifelines...")
    subprocess.check_call([sys.executable, "-m", "pip", "install", "lifelines",␣
 ↪"-q"])
    from lifelines import KaplanMeierFitter, CoxPHFitter
    from lifelines.utils import concordance_index
```

[34]:
```
RANDOM_STATE = 42

# Turn sampling on only if runtime becomes annoying on your machine.
USE_SAMPLE = False
```

```
SAMPLE_SIZE = 250_000

SHORT_HORIZON = 12
LONG_HORIZON = 24

TRAIN_END = "2014-12-31"
VALID_START, VALID_END = "2015-01-01", "2015-12-31"
TEST_START, TEST_END = "2016-01-01", "2019-12-31"

DATA_DIR = Path("data")
RAW_DIR = DATA_DIR / "raw"
OUTPUT_DIR = DATA_DIR / "simplification_PD"
OUTPUT_DIR.mkdir(parents=True, exist_ok=True)

print("Output directory:", OUTPUT_DIR.resolve())
```

```
Output directory:
/Users/pattonpatton/Desktop/RiskProject/Risk_Management_Project/Time
Based/data/simplification_PD
```

0.1 2. Find and load the LendingClub file

[35]:
```
def find_accepted_file():
    candidates = [
        Path("data/raw/lendingclub/accepted_2007_to_2018Q4.csv.gz"),
        Path("data/raw/accepted_2007_to_2018Q4.csv.gz"),
        Path("accepted_2007_to_2018Q4.csv.gz"),
        Path("/mnt/data/accepted_2007_to_2018Q4.csv.gz"),
        Path("/mnt/data/data/raw/lendingclub/accepted_2007_to_2018Q4.csv.gz"),
    ]
    for path in candidates:
        if path.exists():
            return path
    raise FileNotFoundError(
        "Could not find accepted_2007_to_2018Q4.csv.gz. Put it in data/raw/
 ↪lendingclub/ "
        "or edit find_accepted_file()."
    )

accepted_path = find_accepted_file()
print("Using raw file:", accepted_path)
```

```
Using raw file: data/raw/lendingclub/accepted_2007_to_2018Q4.csv.gz
```

[36]:
```
key_columns = [
    "id",
    "loan_status",
    "issue_d",
```

```
    "term",
    "grade",
    "int_rate",
    "loan_amnt",
    "installment",
    "emp_length",
    "home_ownership",
    "annual_inc",
    "verification_status",
    "purpose",
    "dti",
    "delinq_2yrs",
    "inq_last_6mths",
    "open_acc",
    "pub_rec",
    "revol_bal",
    "revol_util",
    "total_acc",
    "mort_acc",
    "pub_rec_bankruptcies",
    "fico_range_low",
    "fico_range_high",
    "earliest_cr_line",
    "last_pymnt_d",
    "last_credit_pull_d",
]

df_raw = pd.read_csv(
    accepted_path,
    usecols=lambda c: c in key_columns,
    low_memory=False,
    compression="gzip",
)

if USE_SAMPLE:
    df_raw = df_raw.sample(min(SAMPLE_SIZE, len(df_raw)),␣
 ↪random_state=RANDOM_STATE).copy()

print("Raw shape:", df_raw.shape)
display(df_raw.head(3))
```

```
Raw shape: (2260701, 28)

         id  loan_amnt        term  int_rate  installment grade emp_length  \
0  68407277     3600.0   36 months     13.99       123.03     C  10+ years
1  68355089    24700.0   36 months     11.99       820.28     C  10+ years
2  68341763    20000.0   60 months     10.78       432.66     B  10+ years
```

```
  home_ownership  annual_inc verification_status  … inq_last_6mths open_acc  \
0       MORTGAGE     55000.0        Not Verified  …            1.0      7.0
1       MORTGAGE     65000.0        Not Verified  …            4.0     22.0
2       MORTGAGE     63000.0        Not Verified  …            0.0      6.0

  pub_rec  revol_bal  revol_util total_acc  last_pymnt_d  last_credit_pull_d  \
0     0.0     2765.0        29.7      13.0      Jan-2019            Mar-2019
1     0.0    21470.0        19.2      38.0      Jun-2016            Mar-2019
2     0.0     7869.0        56.2      18.0      Jun-2017            Mar-2019

   mort_acc  pub_rec_bankruptcies
0       1.0                   0.0
1       4.0                   0.0
2       5.0                   0.0

[3 rows x 28 columns]
```

0.2 3. Clean the loan-level data

[37]:
```
df = df_raw.copy()

for c in ["issue_d", "earliest_cr_line", "last_pymnt_d", "last_credit_pull_d"]:
    df[c] = pd.to_datetime(df[c], format="%b-%Y", errors="coerce")

status = df["loan_status"].fillna("").astype(str)

# Keep the default definition simple and consistent.
default_mask = status.str.contains("Charged Off|Default", case=False,␣
 ↪regex=True)
payoff_mask = status.str.contains("Fully Paid", case=False, regex=False) &␣
 ↪(~default_mask)

df["default_event"] = default_mask.astype(int)
df["payoff_event"] = payoff_mask.astype(int)
df["event_type"] = np.select(
    [df["default_event"] == 1, df["payoff_event"] == 1],
    [1, 2],
    default=0,
)

df["term_num"] = (
    df["term"].astype(str).str.extract(r"(\d+)", expand=False).astype(float)
)

# Strip percentage signs where needed.
df["int_rate"] = (
    df["int_rate"].astype(str).str.replace("%", "", regex=False)
```

```
)
df["int_rate"] = pd.to_numeric(df["int_rate"], errors="coerce")

df["fico_avg"] = (df["fico_range_low"] + df["fico_range_high"]) / 2
df["credit_history_years"] = (
    (df["issue_d"] - df["earliest_cr_line"]).dt.days / 365.25
)

df["end_date"] = df["last_pymnt_d"]
missing_end = df["end_date"].isna()
df.loc[missing_end, "end_date"] = df.loc[missing_end, "last_credit_pull_d"]

def month_diff(start, end):
    return (end.dt.year - start.dt.year) * 12 + (end.dt.month - start.dt.month)

df = df[df["issue_d"].notna()].copy()
df = df[df["end_date"].notna()].copy()

df["duration_months"] = month_diff(df["issue_d"], df["end_date"]).clip(lower=1)

numeric_cols = [
    "int_rate",
    "loan_amnt",
    "installment",
    "annual_inc",
    "dti",
    "delinq_2yrs",
    "inq_last_6mths",
    "open_acc",
    "pub_rec",
    "revol_bal",
    "revol_util",
    "total_acc",
    "mort_acc",
    "pub_rec_bankruptcies",
    "fico_avg",
    "credit_history_years",
    "term_num",
]

for c in numeric_cols:
    df[c] = pd.to_numeric(df[c], errors="coerce")

df_model = df[df["issue_d"] <= pd.Timestamp(TEST_END)].copy()

print("Clean modeling shape:", df_model.shape)
print("Default share:", round(df_model["default_event"].mean(), 4))
```

```
print("Payoff share :", round(df_model["payoff_event"].mean(), 4))
display(df_model[["id", "issue_d", "loan_status", "default_event",␣
↪"payoff_event", "duration_months"]].head())
```

```
Clean modeling shape: (2260668, 36)
Default share: 0.1192
Payoff share : 0.4772

         id    issue_d loan_status  default_event  payoff_event  \
0  68407277 2015-12-01  Fully Paid              0             1
1  68355089 2015-12-01  Fully Paid              0             1
2  68341763 2015-12-01  Fully Paid              0             1
3  66310712 2015-12-01     Current              0             0
4  68476807 2015-12-01  Fully Paid              0             1

   duration_months
0               37
1                6
2               18
3               38
4                7
```

[38]:
```
train_loans = df_model[df_model["issue_d"] <= pd.Timestamp(TRAIN_END)].copy()
valid_loans = df_model[
    (df_model["issue_d"] >= pd.Timestamp(VALID_START)) &
    (df_model["issue_d"] <= pd.Timestamp(VALID_END))
].copy()
test_loans = df_model[
    (df_model["issue_d"] >= pd.Timestamp(TEST_START)) &
    (df_model["issue_d"] <= pd.Timestamp(TEST_END))
].copy()

split_table = pd.DataFrame([
    {"split": "train", "n_loans": len(train_loans), "issue_start":␣
↪train_loans["issue_d"].min(), "issue_end": train_loans["issue_d"].max()},
    {"split": "valid", "n_loans": len(valid_loans), "issue_start":␣
↪valid_loans["issue_d"].min(), "issue_end": valid_loans["issue_d"].max()},
    {"split": "test", "n_loans": len(test_loans), "issue_start":␣
↪test_loans["issue_d"].min(), "issue_end": test_loans["issue_d"].max()},
])
display(split_table)
```

```
   split  n_loans issue_start  issue_end
0  train   466345  2007-06-01 2014-12-01
1  valid   421095  2015-01-01 2015-12-01
2   test  1373228  2016-01-01 2018-12-01
```

0.3 5. Build fixed-horizon labels

[39]:
```
def make_horizon_label(loans, horizon):
    out = loans.copy()
    out["known_horizon_outcome"] = (
        (out["duration_months"] >= horizon) |
        ((out["event_type"].isin([1, 2])) & (out["duration_months"] <= horizon))
    )
    out = out[out["known_horizon_outcome"]].copy()
    out[f"default_within_{horizon}m"] = (
        (out["default_event"] == 1) & (out["duration_months"] <= horizon)
    ).astype(int)
    return out

train_12 = make_horizon_label(train_loans, SHORT_HORIZON)
valid_12 = make_horizon_label(valid_loans, SHORT_HORIZON)
test_12 = make_horizon_label(test_loans, SHORT_HORIZON)

train_24 = make_horizon_label(train_loans, LONG_HORIZON)
valid_24 = make_horizon_label(valid_loans, LONG_HORIZON)
test_24 = make_horizon_label(test_loans, LONG_HORIZON)

label_summary = pd.DataFrame([
    {"dataset": "train_12", "n": len(train_12), "default_rate":␣
 ↪train_12[f"default_within_{SHORT_HORIZON}m"].mean()},
    {"dataset": "valid_12", "n": len(valid_12), "default_rate":␣
 ↪valid_12[f"default_within_{SHORT_HORIZON}m"].mean()},
    {"dataset": "test_12",  "n": len(test_12),  "default_rate":␣
 ↪test_12[f"default_within_{SHORT_HORIZON}m"].mean()},
    {"dataset": "train_24", "n": len(train_24), "default_rate":␣
 ↪train_24[f"default_within_{LONG_HORIZON}m"].mean()},
    {"dataset": "valid_24", "n": len(valid_24), "default_rate":␣
 ↪valid_24[f"default_within_{LONG_HORIZON}m"].mean()},
    {"dataset": "test_24",  "n": len(test_24),  "default_rate":␣
 ↪test_24[f"default_within_{LONG_HORIZON}m"].mean()},
])
display(label_summary)
```

```
    dataset        n  default_rate
0  train_12   466345      0.051022
1  valid_12   421095      0.060556
2   test_12  1016181      0.065078
3  train_24   466345      0.113403
4  valid_24   421095      0.133633
5   test_24   707053      0.151501
```

0.4 6. Features

[40]:
```
model_numeric = [
    "int_rate",
    "loan_amnt",
    "installment",
    "annual_inc",
    "dti",
    "delinq_2yrs",
    "inq_last_6mths",
    "open_acc",
    "pub_rec",
    "revol_bal",
    "revol_util",
    "total_acc",
    "mort_acc",
    "pub_rec_bankruptcies",
    "fico_avg",
    "credit_history_years",
    "term_num",
]

model_categorical = [
    "grade",
    "purpose",
    "home_ownership",
    "emp_length",
    "verification_status",
]

preprocessor = ColumnTransformer(
    transformers=[
        ("num", Pipeline([
            ("imputer", SimpleImputer(strategy="median")),
            ("scaler", StandardScaler()),
        ]), model_numeric),
        ("cat", Pipeline([
            ("imputer", SimpleImputer(strategy="most_frequent")),
            ("onehot", OneHotEncoder(handle_unknown="ignore")),
        ]), model_categorical),
    ]
)
```

0.5 7. Frequency-based models: LPM and Logit

[53]:
```
def fit_frequency_models(train_df, valid_df, test_df, horizon):
    ycol = f"default_within_{horizon}m"
    X_train = train_df[model_numeric + model_categorical]
    y_train = train_df[ycol]
    X_valid = valid_df[model_numeric + model_categorical]
    y_valid = valid_df[ycol]
    X_test = test_df[model_numeric + model_categorical]
    y_test = test_df[ycol]

    lpm = Pipeline([
        ("prep", preprocessor),
        ("model", LinearRegression()),
    ])
    logit = Pipeline([
        ("prep", preprocessor),
        ("model", LogisticRegression(max_iter=2000, C=0.5,␣
 ↪random_state=RANDOM_STATE)),
    ])
    rf_raw = Pipeline([
        ("prep", preprocessor),
        ("model", RandomForestClassifier(
            n_estimators=200,
            max_depth=12,
            min_samples_leaf=50,
            max_features="sqrt",
            class_weight="balanced_subsample",
            max_samples=0.7,
            random_state=RANDOM_STATE,
            n_jobs=-1,
        )),
    ])
    rf_raw.fit(X_train, y_train)
    rf = CalibratedClassifierCV(estimator=rf_raw, method="sigmoid", cv=3)

    lpm.fit(X_train, y_train)
    logit.fit(X_train, y_train)
    rf.fit(X_train, y_train)

    lpm_valid_raw = lpm.predict(X_valid)
    lpm_test_raw = lpm.predict(X_test)
    lpm_valid = np.clip(lpm_valid_raw, 0, 1)
    lpm_test = np.clip(lpm_test_raw, 0, 1)

    logit_valid = logit.predict_proba(X_valid)[:, 1]
    logit_test = logit.predict_proba(X_test)[:, 1]
```

```python
rf_valid = rf.predict_proba(X_valid)[:, 1]
rf_test = rf.predict_proba(X_test)[:, 1]

results = pd.DataFrame([
    {
        "model": f"LPM_{horizon}m",
        "valid_roc_auc": roc_auc_score(y_valid, lpm_valid),
        "valid_pr_auc": average_precision_score(y_valid, lpm_valid),
        "valid_brier": brier_score_loss(y_valid, lpm_valid),
        "test_roc_auc": roc_auc_score(y_test, lpm_test),
        "test_pr_auc": average_precision_score(y_test, lpm_test),
        "test_brier": brier_score_loss(y_test, lpm_test),
    },
    {
        "model": f"Logit_{horizon}m",
        "valid_roc_auc": roc_auc_score(y_valid, logit_valid),
        "valid_pr_auc": average_precision_score(y_valid, logit_valid),
        "valid_brier": brier_score_loss(y_valid, logit_valid),
        "test_roc_auc": roc_auc_score(y_test, logit_test),
        "test_pr_auc": average_precision_score(y_test, logit_test),
        "test_brier": brier_score_loss(y_test, logit_test),
    },
    {
        "model": f"RF_{horizon}m",
        "valid_roc_auc": roc_auc_score(y_valid, rf_valid),
        "valid_pr_auc": average_precision_score(y_valid, rf_valid),
        "valid_brier": brier_score_loss(y_valid, rf_valid),
        "test_roc_auc": roc_auc_score(y_test, rf_test),
        "test_pr_auc": average_precision_score(y_test, rf_test),
        "test_brier": brier_score_loss(y_test, rf_test),
    },
])

payload = {
    "lpm": lpm,
    "logit": logit,
    "rf": rf,
    "rf_raw": rf_raw,
    "valid_pred_lpm": lpm_valid,
    "test_pred_lpm": lpm_test,
    "valid_pred_logit": logit_valid,
    "test_pred_logit": logit_test,
    "valid_pred_rf": rf_valid,
    "test_pred_rf": rf_test,
    "y_valid": y_valid,
    "y_test": y_test,
```

```
        "results": results,
    }
    return payload

freq12 = fit_frequency_models(train_12, valid_12, test_12, SHORT_HORIZON)
freq24 = fit_frequency_models(train_24, valid_24, test_24, LONG_HORIZON)

frequency_results = pd.concat(
    [
        freq12["results"].assign(horizon_months=SHORT_HORIZON),
        freq24["results"].assign(horizon_months=LONG_HORIZON),
    ],
    ignore_index=True,
 )

display(frequency_results)
```

```
       model  valid_roc_auc  valid_pr_auc  valid_brier  test_roc_auc  \
0    LPM_12m       0.711923      0.136887     0.055189      0.698183
1  Logit_12m       0.711386      0.134783     0.055190      0.698867
2     RF_12m       0.714953      0.135385     0.055145      0.704116
3    LPM_24m       0.698317      0.255587     0.109941      0.692691
4  Logit_24m       0.697765      0.254254     0.109875      0.693415
5     RF_24m       0.700539      0.255529     0.109768      0.695883

   test_pr_auc  test_brier  horizon_months
0     0.139254    0.059315              12
1     0.137355    0.059394              12
2     0.140068    0.059254              12
3     0.279992    0.122923              24
4     0.280805    0.122609              24
5     0.282027    0.123107              24
```

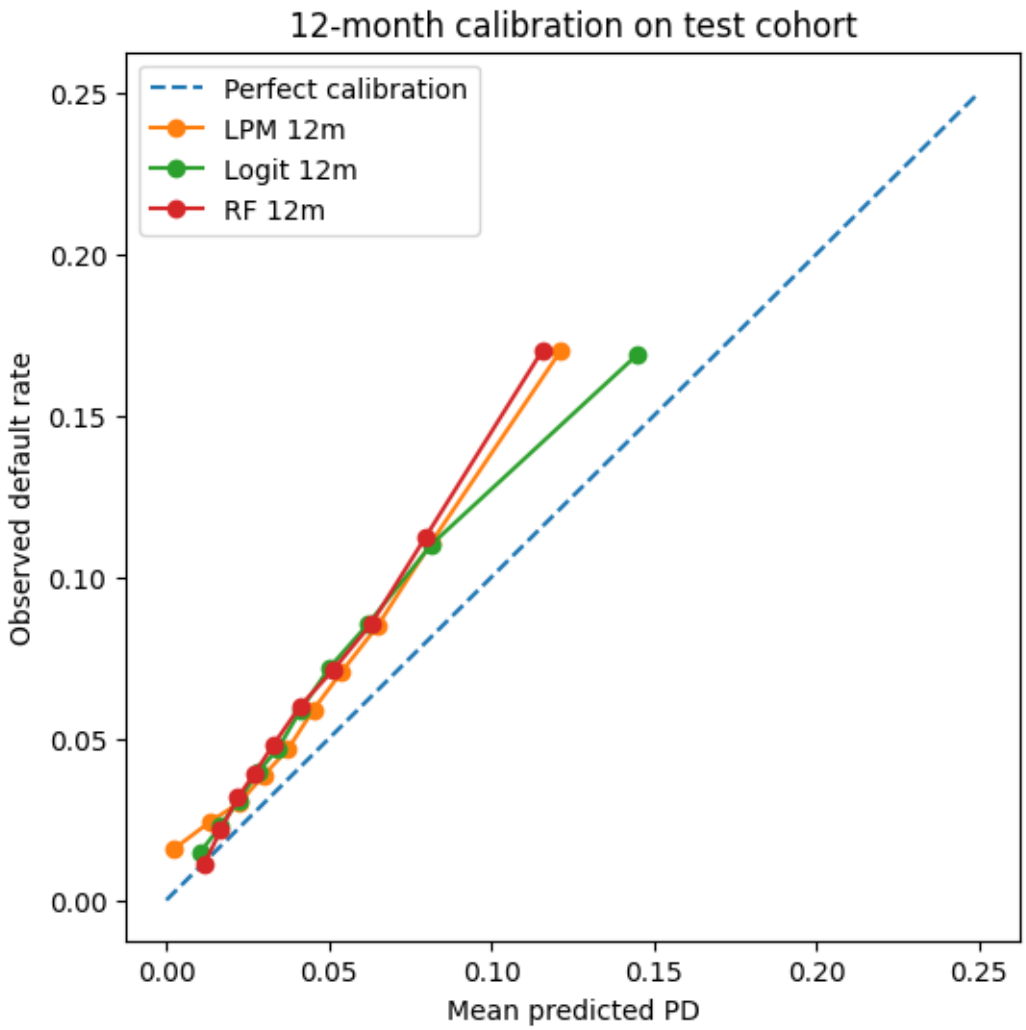
0.5.1 7.1 Accuracy visuals for the frequency models

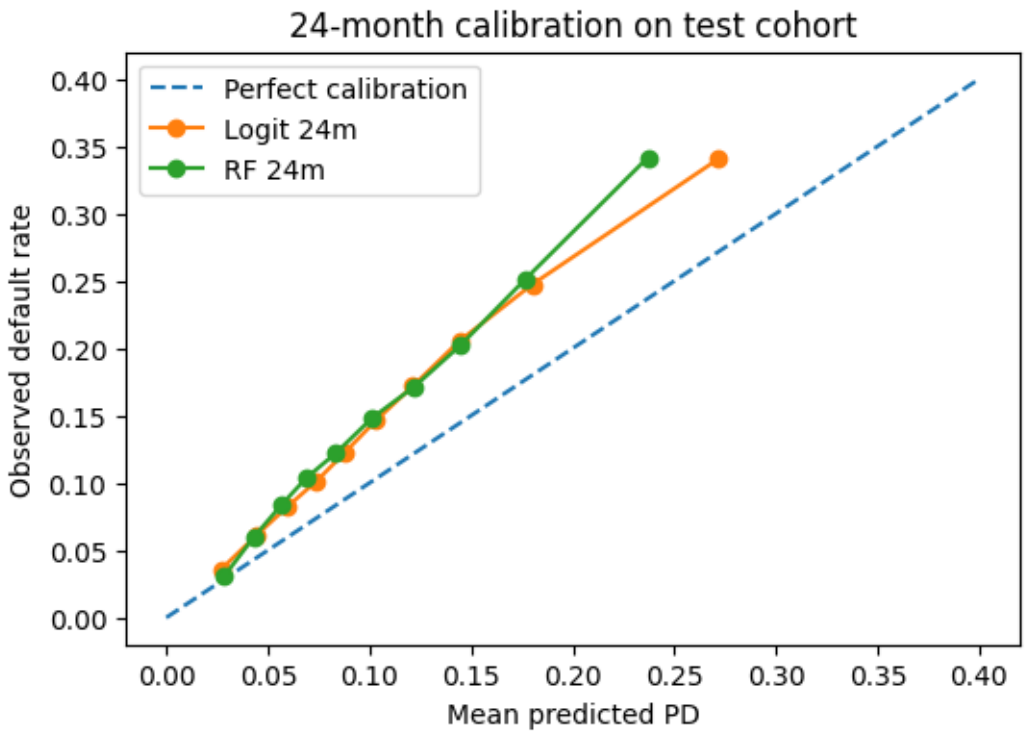
[54]:
```
# 12m calibration: LPM vs Logit vs RF
prob_true_lpm_12, prob_pred_lpm_12 = calibration_curve(freq12["y_test"],␣
 ↪freq12["test_pred_lpm"], n_bins=10, strategy="quantile")
prob_true_logit_12, prob_pred_logit_12 = calibration_curve(freq12["y_test"],␣
 ↪freq12["test_pred_logit"], n_bins=10, strategy="quantile")
prob_true_rf_12, prob_pred_rf_12 = calibration_curve(freq12["y_test"],␣
 ↪freq12["test_pred_rf"], n_bins=10, strategy="quantile")

plt.figure(figsize=(6, 6))
plt.plot([0, 0.25], [0, 0.25], linestyle="--", label="Perfect calibration")
plt.plot(prob_pred_lpm_12, prob_true_lpm_12, marker="o", label="LPM 12m")
plt.plot(prob_pred_logit_12, prob_true_logit_12, marker="o", label="Logit 12m")
```

```
plt.plot(prob_pred_rf_12, prob_true_rf_12, marker="o", label="RF 12m")
plt.xlabel("Mean predicted PD")
plt.ylabel("Observed default rate")
plt.title("12-month calibration on test cohort")
plt.legend()
plt.show()

# 24m calibration: compare all fixed-horizon models against Cox later
prob_true_logit_24, prob_pred_logit_24 = calibration_curve(freq24["y_test"],␣
 ↪freq24["test_pred_logit"], n_bins=10, strategy="quantile")
prob_true_rf_24, prob_pred_rf_24 = calibration_curve(freq24["y_test"],␣
 ↪freq24["test_pred_rf"], n_bins=10, strategy="quantile")
plt.figure(figsize=(6, 4))
plt.plot([0, 0.40], [0, 0.40], linestyle="--", label="Perfect calibration")
plt.plot(prob_pred_logit_24, prob_true_logit_24, marker="o", label="Logit 24m")
plt.plot(prob_pred_rf_24, prob_true_rf_24, marker="o", label="RF 24m")
plt.xlabel("Mean predicted PD")
plt.ylabel("Observed default rate")
plt.title("24-month calibration on test cohort")
plt.legend()
plt.show()
```


12-month calibration on test cohort
Perfect calibration
LPM 12m
Logit 12m
RF 12m
Observed default rate
Mean predicted PD
0.00
0.05
0.10
0.15
0.20
0.25



0.5.2 7.2 Random forest feature importance

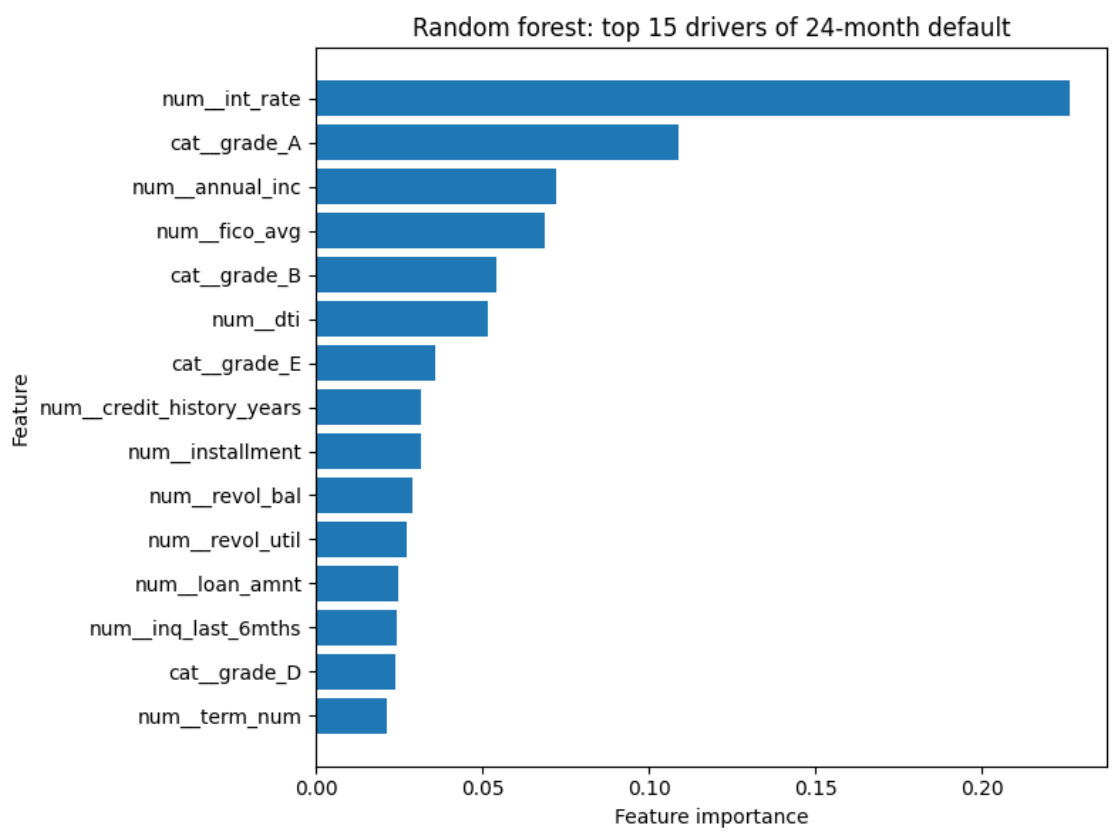
The forest is a benchmark rather than the main PD model, so this view helps check whether it is picking up the same broad credit signals as the Cox fit.

[55]:
```
rf_feature_names_24 = freq24["rf_raw"].named_steps["prep"].
 ↪get_feature_names_out()
rf_feature_importance_24 = pd.DataFrame({
    "feature": rf_feature_names_24,
    "importance": freq24["rf_raw"].named_steps["model"].feature_importances_,
}).sort_values("importance", ascending=False)

display(rf_feature_importance_24.head(20))

plt.figure(figsize=(8, 6))
top_rf_importance = rf_feature_importance_24.head(15).sort_values("importance")
plt.barh(top_rf_importance["feature"], top_rf_importance["importance"])
plt.xlabel("Feature importance")
plt.ylabel("Feature")
plt.title("Random forest: top 15 drivers of 24-month default")
plt.tight_layout()
plt.show()
```

	feature	importance
0	num__int_rate	0.226227
17	cat__grade_A	0.108956
3	num__annual_inc	0.072179
14	num__fico_avg	0.068621
18	cat__grade_B	0.054399
4	num__dti	0.051618
21	cat__grade_E	0.035923
15	num__credit_history_years	0.031596
2	num__installment	0.031514
9	num__revol_bal	0.029158
10	num__revol_util	0.027087
1	num__loan_amnt	0.024827
6	num__inq_last_6mths	0.024119
20	cat__grade_D	0.024065
16	num__term_num	0.021414
19	cat__grade_C	0.020724
22	cat__grade_F	0.020656
11	num__total_acc	0.018036
12	num__mort_acc	0.016989
7	num__open_acc	0.015036



0.6 8. Cox PH as the time-to-default extension

[44]:
```
cox_numeric = model_numeric.copy()
cox_categorical = model_categorical.copy()

def prepare_cox_design(train_df, eval_df):
    train_term = train_df[["id", "duration_months", "default_event"] +␣
 ↪cox_numeric + cox_categorical].copy()
    eval_term = eval_df[["id", "duration_months", "default_event"] +␣
 ↪cox_numeric + cox_categorical].copy()

    for frame in [train_term, eval_term]:
        for c in cox_numeric + ["duration_months", "default_event"]:
            frame[c] = pd.to_numeric(frame[c], errors="coerce")
        for c in cox_categorical:
            frame[c] = frame[c].astype("string").fillna("Missing")

    train_term = train_term.dropna(subset=["duration_months", "default_event"]).
 ↪copy()
    eval_term = eval_term.dropna(subset=["duration_months", "default_event"]).
 ↪copy()

    train_medians = train_term[cox_numeric].median(numeric_only=True)
    train_term[cox_numeric] = train_term[cox_numeric].fillna(train_medians)
    eval_term[cox_numeric] = eval_term[cox_numeric].fillna(train_medians)

    train_dm = pd.get_dummies(train_term, columns=cox_categorical,␣
 ↪drop_first=True)
    eval_dm = pd.get_dummies(eval_term, columns=cox_categorical,␣
 ↪drop_first=True)

    feature_cols = [c for c in train_dm.columns if c not in ["id",␣
 ↪"duration_months", "default_event"]]

    for c in feature_cols:
        if c not in eval_dm.columns:
            eval_dm[c] = 0

    extra_eval = [c for c in eval_dm.columns if c not in ["id",␣
 ↪"duration_months", "default_event"] + feature_cols]
    if extra_eval:
        eval_dm = eval_dm.drop(columns=extra_eval)

    train_dm = train_dm[["id", "duration_months", "default_event"] +␣
 ↪feature_cols].copy()
    eval_dm = eval_dm[["id", "duration_months", "default_event"] +␣
 ↪feature_cols].copy()
```

```
    for frame in [train_dm, eval_dm]:
        bool_cols = frame.select_dtypes(include=["bool"]).columns
        if len(bool_cols) > 0:
            frame[bool_cols] = frame[bool_cols].astype(int)
        frame[feature_cols] = frame[feature_cols].astype(float)
        frame["duration_months"] = frame["duration_months"].astype(float)
        frame["default_event"] = frame["default_event"].astype(int)

    nunique = train_dm[feature_cols].nunique(dropna=False)
    constant_cols = nunique[nunique <= 1].index.tolist()
    if constant_cols:
        feature_cols = [c for c in feature_cols if c not in constant_cols]
        train_dm = train_dm[["id", "duration_months", "default_event"] +␣
 ↪feature_cols].copy()
        eval_dm = eval_dm[["id", "duration_months", "default_event"] +␣
 ↪feature_cols].copy()

    return train_dm, eval_dm, feature_cols

cox_train, cox_valid, cox_features = prepare_cox_design(train_loans,␣
 ↪valid_loans)
_, cox_test, _ = prepare_cox_design(train_loans, test_loans)

cph = CoxPHFitter(penalizer=0.1)
cph.fit(
    cox_train.drop(columns=["id"]),
    duration_col="duration_months",
    event_col="default_event",
    show_progress=False,
)

valid_risk = cph.predict_partial_hazard(cox_valid[cox_features]).values.ravel()
test_risk = cph.predict_partial_hazard(cox_test[cox_features]).values.ravel()

cox_c_valid = concordance_index(
    cox_valid["duration_months"],
    -valid_risk,
    cox_valid["default_event"],
)
cox_c_test = concordance_index(
    cox_test["duration_months"],
    -test_risk,
    cox_test["default_event"],
)

cox_summary = pd.DataFrame([{
```

```
    "model": "Cox_PH_time_to_default",
    "valid_c_index": cox_c_valid,
    "test_c_index": cox_c_test,
    "n_train": len(cox_train),
    "n_valid": len(cox_valid),
    "n_test": len(cox_test),
}])

display(cox_summary)
```

```
                    model  valid_c_index  test_c_index  n_train  n_valid  \
0  Cox_PH_time_to_default       0.685673      0.693352   466345   421095

    n_test
0  1373228
```

0.6.1 8.1 The strongest Cox hazard ratios

[45]:
```
cox_coef_view = cph.summary[["coef", "exp(coef)", "se(coef)", "z", "p"]].
 ↪sort_values("exp(coef)", ascending=False)
display(cox_coef_view.head(20))
```

```
                                       coef  exp(coef)  se(coef)          z  \
covariate
grade_G                            0.327748   1.387840  0.027008  12.135345
purpose_small_business             0.289166   1.335313  0.021408  13.507466
grade_F                            0.266647   1.305580  0.015510  17.191774
grade_E                            0.245254   1.277946  0.010704  22.912775
grade_D                            0.184431   1.202534  0.008228  22.415839
emp_length_Missing                 0.176933   1.193551  0.013567  13.041089
purpose_educational                0.174224   1.190323  0.090512   1.924884
purpose_renewable_energy           0.129382   1.138124  0.101379   1.276221
home_ownership_OTHER               0.126533   1.134887  0.136350   0.927997
grade_C                            0.092935   1.097390  0.007151  12.995780
purpose_moving                     0.077684   1.080781  0.034387   2.259090
purpose_medical                    0.070724   1.073285  0.028658   2.467877
inq_last_6mths                     0.066382   1.068635  0.002498  26.574365
purpose_house                      0.062333   1.064316  0.040383   1.543523
home_ownership_RENT                0.061264   1.063179  0.007255   8.444225
pub_rec_bankruptcies               0.058406   1.060145  0.009286   6.289333
verification_status_Source Verified  0.054143   1.055636  0.006574   8.236564
int_rate                           0.039760   1.040561  0.000828  48.020705
purpose_debt_consolidation         0.039603   1.040398  0.007054   5.614187
emp_length_< 1 year                0.026845   1.027209  0.010994   2.441776

                                            p
covariate
grade_G                          6.862024e-34
```

```
purpose_small_business                1.413014e-41
grade_F                               3.060243e-66
grade_E                              3.465535e-116
grade_D                              2.757953e-111
emp_length_Missing                    7.142965e-39
purpose_educational                   5.424384e-02
purpose_renewable_energy              2.018775e-01
home_ownership_OTHER                  3.534089e-01
grade_C                               1.292834e-38
purpose_moving                        2.387778e-02
purpose_medical                       1.359171e-02
inq_last_6mths                       1.343248e-155
purpose_house                         1.227038e-01
home_ownership_RENT                   3.060689e-17
pub_rec_bankruptcies                  3.188323e-10
verification_status_Source Verified   1.772262e-16
int_rate                              0.000000e+00
purpose_debt_consolidation            1.974882e-08
emp_length_< 1 year                   1.461523e-02
```

0.6.2 8.2 Supported-horizon Cox curve check

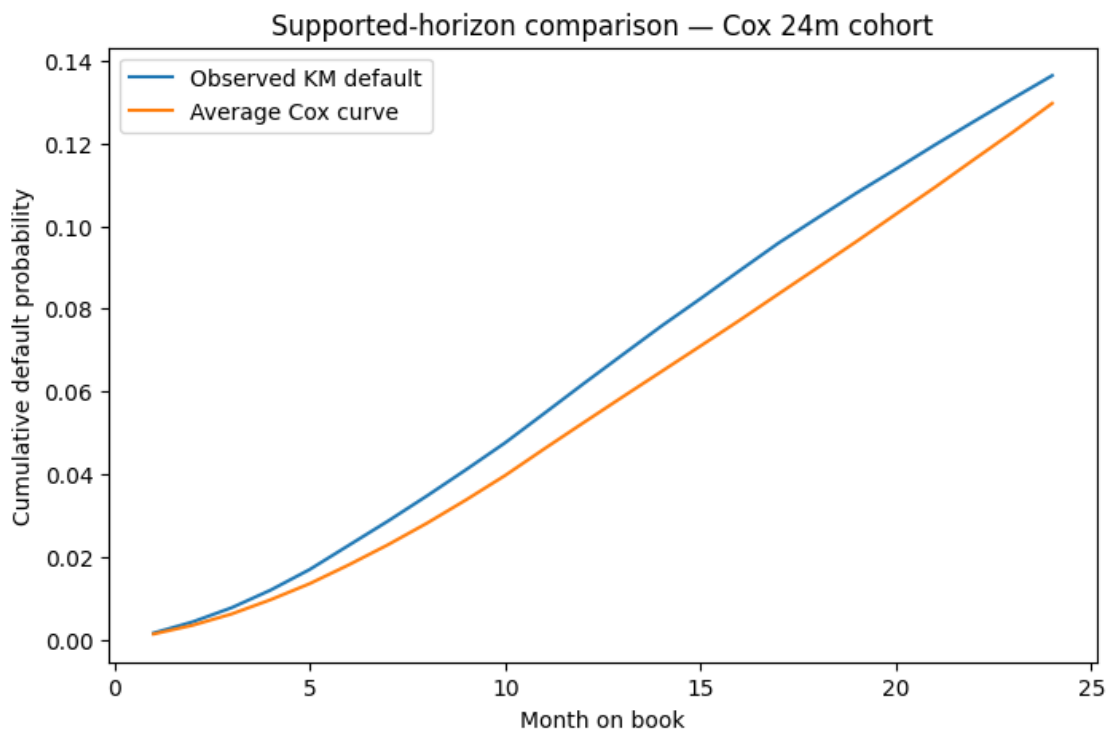
[46]:
```
def km_curve(loans, max_horizon=None):
    kmf = KaplanMeierFitter()
    kmf.fit(
        durations=loans["duration_months"],
        event_observed=loans["default_event"],
    )
    surv = kmf.survival_function_.reset_index()
    surv.columns = ["month_on_book", "survival"]
    surv["month_on_book"] = surv["month_on_book"].astype(float).round().
 ↪astype(int)
    surv["cum_default_km"] = 1 - surv["survival"]
    if max_horizon is not None:
        surv = surv[surv["month_on_book"] <= max_horizon].copy()
    return surv

km_test = km_curve(test_loans, max_horizon=LONG_HORIZON)

surv_pred_test = cph.predict_survival_function(cox_test[cox_features])
avg_surv = surv_pred_test.mean(axis=1).reset_index()
avg_surv.columns = ["month_on_book", "survival_cox_avg"]
avg_surv["month_on_book"] = avg_surv["month_on_book"].astype(float).round().
 ↪astype(int)
avg_surv = avg_surv.groupby("month_on_book",␣
 ↪as_index=False)["survival_cox_avg"].last()
avg_surv["cum_default_cox_avg"] = 1 - avg_surv["survival_cox_avg"]
```

```
compare_curve = km_test.merge(
    avg_surv[["month_on_book", "cum_default_cox_avg"]],
    on="month_on_book",
    how="inner",
)

plt.figure(figsize=(8, 5))
plt.plot(compare_curve["month_on_book"], compare_curve["cum_default_km"],␣
 ↪label="Observed KM default")
plt.plot(compare_curve["month_on_book"], compare_curve["cum_default_cox_avg"],␣
 ↪label="Average Cox curve")
plt.xlabel("Month on book")
plt.ylabel("Cumulative default probability")
plt.title(f"Supported-horizon comparison - Cox {LONG_HORIZON}m cohort")
plt.legend()
plt.show()
```



0.7 9. Convert Cox into a simple 24-month PD and compare it with 24-month logit

[47]:
```
# 24m PD from Cox on the validation/test sets
cox_valid_24 = cph.predict_survival_function(cox_valid[cox_features],␣
 ↪times=[LONG_HORIZON]).T
cox_valid_24.columns = ["survival_24m"]
cox_valid_24["pd_24m_cox"] = 1 - cox_valid_24["survival_24m"]
cox_valid_24 = cox_valid_24.reset_index(drop=True)
cox_valid_24["id"] = cox_valid["id"].values

cox_test_24 = cph.predict_survival_function(cox_test[cox_features],␣
 ↪times=[LONG_HORIZON]).T
cox_test_24.columns = ["survival_24m"]
cox_test_24["pd_24m_cox"] = 1 - cox_test_24["survival_24m"]
cox_test_24 = cox_test_24.reset_index(drop=True)
cox_test_24["id"] = cox_test["id"].values

valid24_compare = valid_24[["id", f"default_within_{LONG_HORIZON}m"]].merge(
    cox_valid_24[["id", "pd_24m_cox"]],
    on="id",
    how="inner",
)
test24_compare = test_24[["id", f"default_within_{LONG_HORIZON}m"]].merge(
    cox_test_24[["id", "pd_24m_cox"]],
    on="id",
    how="inner",
)

valid24_compare["pd_24m_logit"] = freq24["valid_pred_logit"]
test24_compare["pd_24m_logit"] = freq24["test_pred_logit"]
valid24_compare["pd_24m_rf"] = freq24["valid_pred_rf"]
test24_compare["pd_24m_rf"] = freq24["test_pred_rf"]

y_valid_24 = valid24_compare[f"default_within_{LONG_HORIZON}m"]
y_test_24 = test24_compare[f"default_within_{LONG_HORIZON}m"]

longterm_results = pd.DataFrame([
    {
        "model": "Logit_24m",
        "valid_roc_auc": roc_auc_score(y_valid_24,␣
 ↪valid24_compare["pd_24m_logit"]),
        "valid_pr_auc": average_precision_score(y_valid_24,␣
 ↪valid24_compare["pd_24m_logit"]),
        "valid_brier": brier_score_loss(y_valid_24,␣
 ↪valid24_compare["pd_24m_logit"]),
```

```
        "test_roc_auc": roc_auc_score(y_test_24,␣
↪test24_compare["pd_24m_logit"]),
        "test_pr_auc": average_precision_score(y_test_24,␣
↪test24_compare["pd_24m_logit"]),
        "test_brier": brier_score_loss(y_test_24,␣
↪test24_compare["pd_24m_logit"]),
    },
    {
        "model": "Cox_implied_PD_24m",
        "valid_roc_auc": roc_auc_score(y_valid_24,␣
↪valid24_compare["pd_24m_cox"]),
        "valid_pr_auc": average_precision_score(y_valid_24,␣
↪valid24_compare["pd_24m_cox"]),
        "valid_brier": brier_score_loss(y_valid_24,␣
↪valid24_compare["pd_24m_cox"]),
        "test_roc_auc": roc_auc_score(y_test_24, test24_compare["pd_24m_cox"]),
        "test_pr_auc": average_precision_score(y_test_24,␣
↪test24_compare["pd_24m_cox"]),
        "test_brier": brier_score_loss(y_test_24, test24_compare["pd_24m_cox"]),
    },
    {
        "model": "RF_24m",
        "valid_roc_auc": roc_auc_score(y_valid_24,␣
↪valid24_compare["pd_24m_rf"]),
        "valid_pr_auc": average_precision_score(y_valid_24,␣
↪valid24_compare["pd_24m_rf"]),
        "valid_brier": brier_score_loss(y_valid_24,␣
↪valid24_compare["pd_24m_rf"]),
        "test_roc_auc": roc_auc_score(y_test_24, test24_compare["pd_24m_rf"]),
        "test_pr_auc": average_precision_score(y_test_24,␣
↪test24_compare["pd_24m_rf"]),
        "test_brier": brier_score_loss(y_test_24, test24_compare["pd_24m_rf"]),
    },
])

display(longterm_results)
```

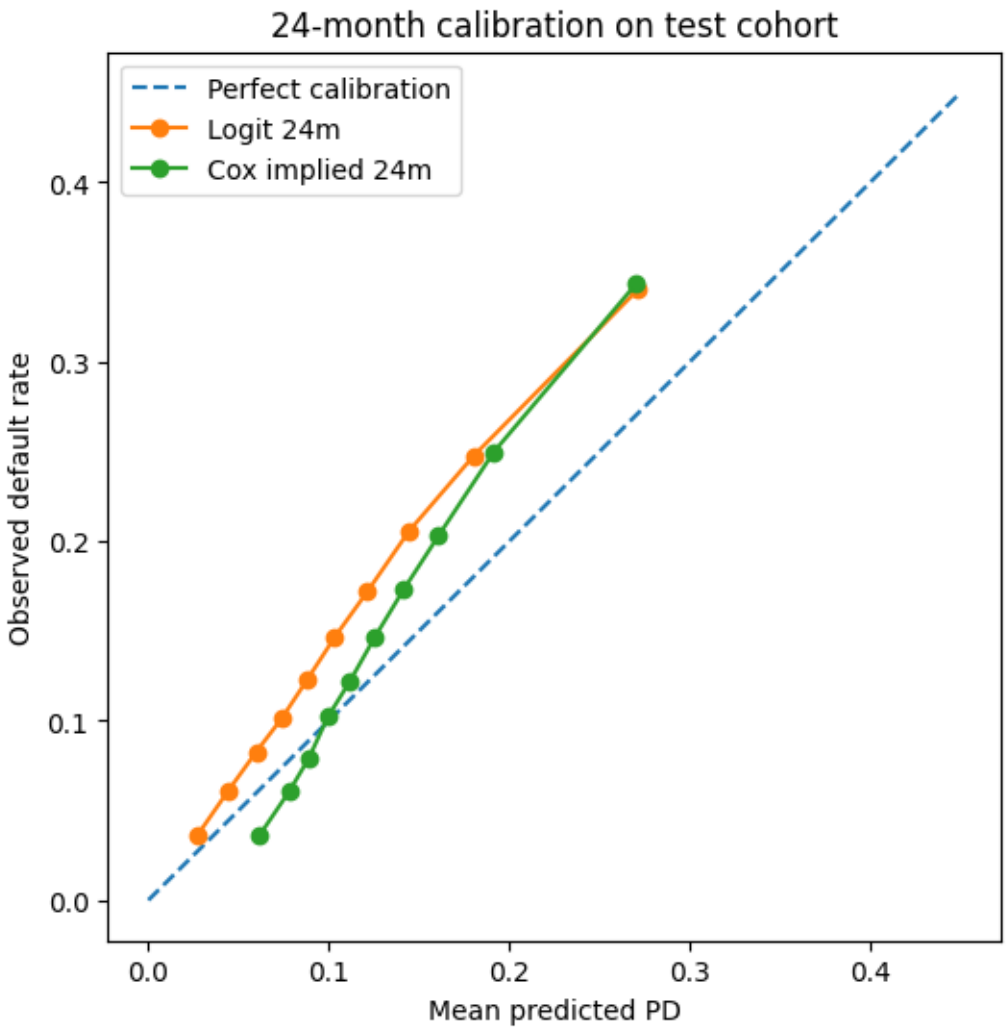
```
                model  valid_roc_auc  valid_pr_auc  valid_brier  test_roc_auc  \
0           Logit_24m       0.697765      0.254254     0.109875      0.693415
1  Cox_implied_PD_24m       0.699143      0.257015     0.109599      0.695005
2              RF_24m       0.700539      0.255529     0.109768      0.695883

   test_pr_auc  test_brier
0     0.280805    0.122609
1     0.283241    0.121670
2     0.282027    0.123107
```

0.7.1 9.1 Calibration: 24m logit vs 24m Cox

[48]:
```
prob_true_logit_24, prob_pred_logit_24 = calibration_curve(
    y_test_24, test24_compare["pd_24m_logit"], n_bins=10, strategy="quantile"
)
prob_true_cox_24, prob_pred_cox_24 = calibration_curve(
    y_test_24, test24_compare["pd_24m_cox"], n_bins=10, strategy="quantile"
)

plt.figure(figsize=(6, 6))
plt.plot([0, 0.45], [0, 0.45], linestyle="--", label="Perfect calibration")
plt.plot(prob_pred_logit_24, prob_true_logit_24, marker="o", label="Logit 24m")
plt.plot(prob_pred_cox_24, prob_true_cox_24, marker="o", label="Cox implied␣
↪24m")
plt.xlabel("Mean predicted PD")
plt.ylabel("Observed default rate")
plt.title("24-month calibration on test cohort")
plt.legend()
plt.show()
```



0.8 10. Choose the simplest long-term model

[49]:
```
lt = longterm_results.set_index("model")
rf_margin = 0.001

rf_brier = lt.loc["RF_24m", "valid_brier"]
cox_brier = lt.loc["Cox_implied_PD_24m", "valid_brier"]
logit_brier = lt.loc["Logit_24m", "valid_brier"]

rf_is_best = rf_brier <= min(cox_brier, logit_brier) - rf_margin

if rf_is_best:
    chosen_longterm_model = "RF_24m"
elif cox_brier <= logit_brier - rf_margin:
    chosen_longterm_model = "Cox_implied_PD_24m"
```

```
else:
    chosen_longterm_model = "Logit_24m"

selection_table = longterm_results.copy()
selection_table["chosen_for_simple_longterm"] = selection_table["model"].
 ↪eq(chosen_longterm_model)
display(selection_table)

print("Chosen simple long-term PD model:", chosen_longterm_model)
```

```
                model  valid_roc_auc  valid_pr_auc  valid_brier  test_roc_auc  \
0           Logit_24m       0.697765      0.254254     0.109875      0.693415
1  Cox_implied_PD_24m       0.699143      0.257015     0.109599      0.695005
2              RF_24m       0.700539      0.255529     0.109768      0.695883

   test_pr_auc  test_brier  chosen_for_simple_longterm
0     0.280805    0.122609                        True
1     0.283241    0.121670                       False
2     0.282027    0.123107                       False

Chosen simple long-term PD model: Logit_24m
```

0.9 11. Refit final production-style models on train + validation

This gives teammates one clean set of PD outputs.

[50]:
```
# Combine train+valid for final fitting
tv_loans = pd.concat([train_loans, valid_loans], ignore_index=True)
tv_12 = pd.concat([train_12, valid_12], ignore_index=True)
tv_24 = pd.concat([train_24, valid_24], ignore_index=True)

# Final 12m logit
X_tv_12 = tv_12[model_numeric + model_categorical]
y_tv_12 = tv_12[f"default_within_{SHORT_HORIZON}m"]

final_logit_12 = Pipeline([
    ("prep", preprocessor),
    ("model", LogisticRegression(max_iter=2000, C=0.5,␣
 ↪random_state=RANDOM_STATE)),
])
final_logit_12.fit(X_tv_12, y_tv_12)

# Final 24m logit
X_tv_24 = tv_24[model_numeric + model_categorical]
y_tv_24 = tv_24[f"default_within_{LONG_HORIZON}m"]

final_logit_24 = Pipeline([
    ("prep", preprocessor),
```

```
    ("model", LogisticRegression(max_iter=2000, C=0.5,␣
 ↪random_state=RANDOM_STATE)),
])
final_logit_24.fit(X_tv_24, y_tv_24)

# Final RF models
final_rf_raw_12 = Pipeline([
    ("prep", preprocessor),
    ("model", RandomForestClassifier(
        n_estimators=200,
        max_depth=12,
        min_samples_leaf=50,
        max_features="sqrt",
        class_weight="balanced_subsample",
        max_samples=0.7,
        random_state=RANDOM_STATE,
        n_jobs=-1,
    )),
])
final_rf_12 = CalibratedClassifierCV(estimator=final_rf_raw_12,␣
 ↪method="sigmoid", cv=3)
final_rf_12.fit(X_tv_12, y_tv_12)

final_rf_raw_24 = Pipeline([
    ("prep", preprocessor),
    ("model", RandomForestClassifier(
        n_estimators=200,
        max_depth=12,
        min_samples_leaf=50,
        max_features="sqrt",
        class_weight="balanced_subsample",
        max_samples=0.7,
        random_state=RANDOM_STATE,
        n_jobs=-1,
    )),
])
final_rf_24 = CalibratedClassifierCV(estimator=final_rf_raw_24,␣
 ↪method="sigmoid", cv=3)
final_rf_24.fit(X_tv_24, y_tv_24)

# Final Cox
final_cox_train, final_cox_all, final_cox_features =␣
 ↪prepare_cox_design(tv_loans, df_model)
final_cph = CoxPHFitter(penalizer=0.1)
final_cph.fit(
    final_cox_train.drop(columns=["id"]),
    duration_col="duration_months",
```

```
    event_col="default_event",
    show_progress=False,
)
```

[50]: <lifelines.CoxPHFitter: fitted with 887440 total observations, 734375 right-censored observations>

0.10 12. Score the whole modeled portfolio

[51]:
```
pd_output = df_model[[
    "id", "issue_d", "loan_status", "grade", "int_rate", "loan_amnt",
    "annual_inc", "dti", "fico_avg", "term_num"
 ]].copy()

X_all = df_model[model_numeric + model_categorical]
pd_output["pd_12m_logit"] = final_logit_12.predict_proba(X_all)[:, 1]
pd_output["pd_24m_logit"] = final_logit_24.predict_proba(X_all)[:, 1]
pd_output["pd_12m_rf"] = final_rf_12.predict_proba(X_all)[:, 1]
pd_output["pd_24m_rf"] = final_rf_24.predict_proba(X_all)[:, 1]

cox_all_24 = final_cph.
 ↪predict_survival_function(final_cox_all[final_cox_features],␣
 ↪times=[LONG_HORIZON]).T
cox_all_24.columns = ["survival_24m"]
cox_all_24["pd_24m_cox"] = 1 - cox_all_24["survival_24m"]
cox_all_24 = cox_all_24.reset_index(drop=True)
cox_all_24["id"] = final_cox_all["id"].values

pd_output = pd_output.merge(cox_all_24[["id", "pd_24m_cox"]], on="id",␣
 ↪how="left")

if chosen_longterm_model == "Cox_implied_PD_24m":
    pd_output["pd_24m_chosen"] = pd_output["pd_24m_cox"]
elif chosen_longterm_model == "RF_24m":
    pd_output["pd_24m_chosen"] = pd_output["pd_24m_rf"]
else:
    pd_output["pd_24m_chosen"] = pd_output["pd_24m_logit"]

pd_output["chosen_longterm_model"] = chosen_longterm_model

display(pd_output.head())

pd_output.to_csv(OUTPUT_DIR / "pd_predictions_for_teammates.csv", index=False)
frequency_results.to_csv(OUTPUT_DIR / "frequency_model_metrics.csv",␣
 ↪index=False)
longterm_results.to_csv(OUTPUT_DIR / "longterm_model_comparison.csv",␣
 ↪index=False)
```

```python
cox_summary.to_csv(OUTPUT_DIR / "cox_time_to_default_summary.csv", index=False)

print("Saved:")
print("-", OUTPUT_DIR / "pd_predictions_for_teammates.csv")
print("-", OUTPUT_DIR / "frequency_model_metrics.csv")
print("-", OUTPUT_DIR / "longterm_model_comparison.csv")
print("-", OUTPUT_DIR / "cox_time_to_default_summary.csv")
```

```
         id    issue_d loan_status grade  int_rate  loan_amnt  annual_inc  \
0  68407277 2015-12-01  Fully Paid     C     13.99     3600.0     55000.0
1  68355089 2015-12-01  Fully Paid     C     11.99    24700.0     65000.0
2  68341763 2015-12-01  Fully Paid     B     10.78    20000.0     63000.0
3  66310712 2015-12-01     Current     C     14.85    35000.0    110000.0
4  68476807 2015-12-01  Fully Paid     F     22.45    10400.0    104433.0

     dti  fico_avg  term_num  pd_12m_logit  pd_24m_logit  pd_12m_rf  \
0   5.91     677.0      36.0      0.043247      0.102377   0.047472
1  16.06     717.0      36.0      0.084606      0.190163   0.053313
2  10.78     697.0      60.0      0.019489      0.054399   0.020744
3  17.06     787.0      60.0      0.038070      0.091440   0.046853
4  25.37     697.0      60.0      0.142994      0.235424   0.104308

   pd_24m_rf  pd_24m_cox  pd_24m_chosen chosen_longterm_model
0   0.099604    0.130483       0.102377             Logit_24m
1   0.124430    0.173624       0.190163             Logit_24m
2   0.053914    0.081046       0.054399             Logit_24m
3   0.104345    0.124820       0.091440             Logit_24m
4   0.221449    0.247683       0.235424             Logit_24m

Saved:
- data/simplification_PD/pd_predictions_for_teammates.csv
- data/simplification_PD/frequency_model_metrics.csv
- data/simplification_PD/longterm_model_comparison.csv
- data/simplification_PD/cox_time_to_default_summary.csv
```